Benchmarking Decision Models for LLM Safety Screening

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Abstract

Under one pre-registered protocol, typed decision models and a managed guardrail sit on different miss–false-positive–cost points; on JailbreakBench the guardrail's headline miss is mostly the category crosswalk, not the strength setting; stacks help only where that guardrail still catches harm. We score TypeSafe Jev (`jev-1.13.0`), Cloudflare Clef and Clef-flash, and Amazon Bedrock Guardrails on eleven public prompt-level benchmarks, with pre-registered union, intersection, and cascade stacks on three sets. A harness check reproduced published Jev intervals on six overlapping datasets (`run_id=20261020T033822Z_0f24960e`, PASS); headline metrics use core pack `20261020T040920Z_90c5fd13`. On JailbreakBench the pre-registered Prompt-Attack-only crosswalk yields guardrail miss 0.9800 (bootstrap 95% CI 0.95–1.00) at false-positive rate 0; a post-hoc any-filter rule at medium strength yields miss 0.23 (CI 0.15–0.31) at FPR 0.19 (CI 0.11–0.26). Jev at threshold 0.5 reaches miss 0.03 (CI 0.00–0.07) at FPR 0.34 (CI 0.25–0.44). HarmBench overall miss is 0.2026 for guardrails versus 0.2870 for Jev, but splits reverse by attack type. Jev is the cheapest and fastest arm on this pack ($0.023 per 1,000 prompts; median latency 88.9 ms) and does not win every miss metric.

1 Introduction

Input screening trades missed harm against over-refusal, latency, and cost. Two commercial families implement that layer differently. Typed decision models—TypeSafe Jev TypeSafe AI [2026], Deußer et al. [2026b], Cloudflare Clef, and Clef-flash Cloudflare [2026a,b]—take fixed questions on prompt text and return probabilities or discrete choices. Amazon Bedrock Guardrails Amazon Web Services [2024] scores input through `ApplyGuardrail` and maps ordinal bands on six content filters to block decisions. Prior Jev benchmarks Deußer et al. [2026b] do not place a managed guardrail on the same rows under harmonized miss and false-positive definitions.

Under one pre-registered protocol, typed decision models and a managed guardrail sit on different miss–false-positive–cost points; on JailbreakBench the guardrail's headline miss is mostly the category crosswalk, not the strength setting; stacks help only where that guardrail still catches harm.

We evaluate four arms on eleven public prompt-only datasets (safety sets plus GoEmotions and Emotion for emotionally loaded non-harm text). We report tuned decision-model thresholds and scores at 0.5, guardrail metrics at medium strength under a dataset-to-filter crosswalk fixed before test scoring, and pre-registered stacks (union, intersection, cascade) from cached predictions. We do not measure downstream chat-model compliance or user outcomes in live sessions.

Before the full pack we reran the public Jev harness (`20261020T033822Z_0f24960e`, PASS). Headline numbers below use `20261020T040920Z_90c5fd13`. Section 2 situates prior benchmarks; Sections 3–5 describe arms, data, and metrics; Section 6 reports empirical findings; Sections 7–10 interpret deployment trade-offs.

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2 Related Work

Public safety benchmarks—ToxiGen Hartvigsen et al. [2022], OpenAI Moderation Markov et al. [2023], ToxicChat Lin et al. [2023], prompt injections deepset [2024], WildGuardMix Han et al. [2024], Do-Not-Answer Wang et al. [2024], XSTest Röttger et al. [2024], HarmBench Mazeika et al. [2024], and JailbreakBench Chao et al. [2024]—were built for different model interfaces and primary metrics. Deußer et al. benchmark Jev zero-shot on many of these tasks with a shared harness Deußer et al. [2026b,a] but without Bedrock Guardrails or stack policies on the same prompts. Clef and Clef-flash share the Jev request shape Cloudflare [2026a,b]; Bedrock documents ordinal filter bands rather than calibrated probabilities Amazon Web Services [2024]. Calibration and selective prediction tools apply to probabilistic arms Guo et al. [2017], Geifman and El-Yaniv [2017]; guardrail deployments still require an explicit label-to-filter crosswalk. We add Clef, Clef-flash, guardrails, stacks, and unified miss/FPR/detection semantics on one eleven-dataset pack.

3 Systems Under Test

Four screening arms score the same prompts: Jev `jev-1.13.0` TypeSafe AI [2026], Clef and Clef-flash on Workers AI Cloudflare [2026a,b], and Bedrock Guardrails (six input content filters, detect-only, region `us-east-2`) Amazon Web Services [2024]. Typed models return harm probabilities from versioned question templates; guardrails return ordinal bands recomputed offline at low, medium, or high strength (medium for headlines). A pre-registered crosswalk maps dataset labels to filter subsets; JailbreakBench harmful goals map to Prompt Attack only in registered results. Union blocks if either arm blocks; intersection requires both; cascade defers scores between development-tuned cutoffs to the medium-strength guardrail bit. Stacks read cached predictions only and add no API calls.

4 Datasets

Eleven Hugging Face datasets supply prompt-level labels: nine safety or jailbreak corpora plus GoEmotions and Emotion Hartvigsen et al. [2022], Markov et al. [2023], Lin et al. [2023], deepset [2024], Han et al. [2024], Wang et al. [2024], Röttger et al. [2024], Mazeika et al. [2024], Chao et al. [2024], Demszky et al. [2020], Saravia et al. [2018]. We score user text only, never model replies. Development pools fit thresholds (maximize harmful-class F1 with benign FPR $\leq 5\%$ on labeled benign rows, else 0.5); JailbreakBench stays at 0.5 because its labeled development pool has fewer than 200 rows. HarmBench merges DirectRequest ($n = 320$ test) and HumanJailbreaks ($n = 1600$ test) for overall miss. JailbreakBench balances 100 harmful and 100 benign goals. Do-Not-Answer reports detection rate and risk-area aggregates without reproducing item text. GoEmotions and Emotion use empty guardrail crosswalks (miss rate 1.0 means all rows pass the guardrail layer under our block definition). Full row counts and split rules are in the appendix.

5 Methods

Each loader emits stable example ids, gold labels, and optional benign-known flags. Decision models score versioned question templates; guardrails store `ApplyGuardrail` bands in a SQLite cache keyed by arm, template hash, and example id. Headline metrics use core pack `20261020T040920Z_90c5fd13`; harness validation used `20261020T033822Z_0f24960e`.

Thresholds and guardrails. Tuned thresholds maximize harmful-class F1 on development rows subject to a 5% benign FPR cap; we also report scores at 0.5. Guardrail block bits apply Bedrock strength rules to the max matched band under the pre-registered crosswalk. A post-hoc JailbreakBench variant takes the max band across all six filters without refitting model thresholds; we label it post-hoc throughout.

Stacks and metrics. Pre-registered stacks pair each decision model with the medium-strength crosswalk guardrail on JailbreakBench, XSTest, and WildGuardMix. Miss rate is the share of harmful prompts allowed; FPR is the share of benign-known prompts blocked; Do-Not-Answer uses detection rate on harm-only rows. Bootstrap CIs in the results section are percentile intervals from 1,000 resamples (seed 42) of JailbreakBench harmful and benign rows and of the XSTest Jev operating points. Expected calibration error (ECE-15) summarizes probabilistic arms where noted in the appendix.

6 Results

Headline metrics come from core pack `20261020T040920Z_90c5fd13`; the public Jev harness passed on six overlapping sets (`20261020T033822Z_0f24960e`) before that run. Decision models use development-tuned thresholds unless noted; guardrails use the pre-registered crosswalk at medium strength. Table 1 in the appendix lists the full grid.

6.1 JailbreakBench crosswalk

On JailbreakBench the pre-registered mapping sends harmful goals to Prompt Attack only. At medium strength the guardrail misses 0.9800 of harmful rows (bootstrap mean 0.98, 95% CI 0.95–1.00) with FPR 0. Strength barely moves that point: miss 1.00 at low, 0.98 at medium, and 0.96 at high, with FPR 0 at all three. A post-hoc any-filter rule at medium strength—maximum band across Hate, Insults, Sexual, Violence, Misconduct, and Prompt Attack, not refit on development data—yields miss 0.23 (CI 0.15–0.31) and FPR 0.19 (CI 0.11–0.26). At score threshold 0.5, Jev miss is 0.03 (CI 0.00–0.07) at FPR 0.34 (CI 0.25–0.44); Clef miss 0.01 at FPR 0.54; Clef-flash miss 0.07 at FPR 0.24.

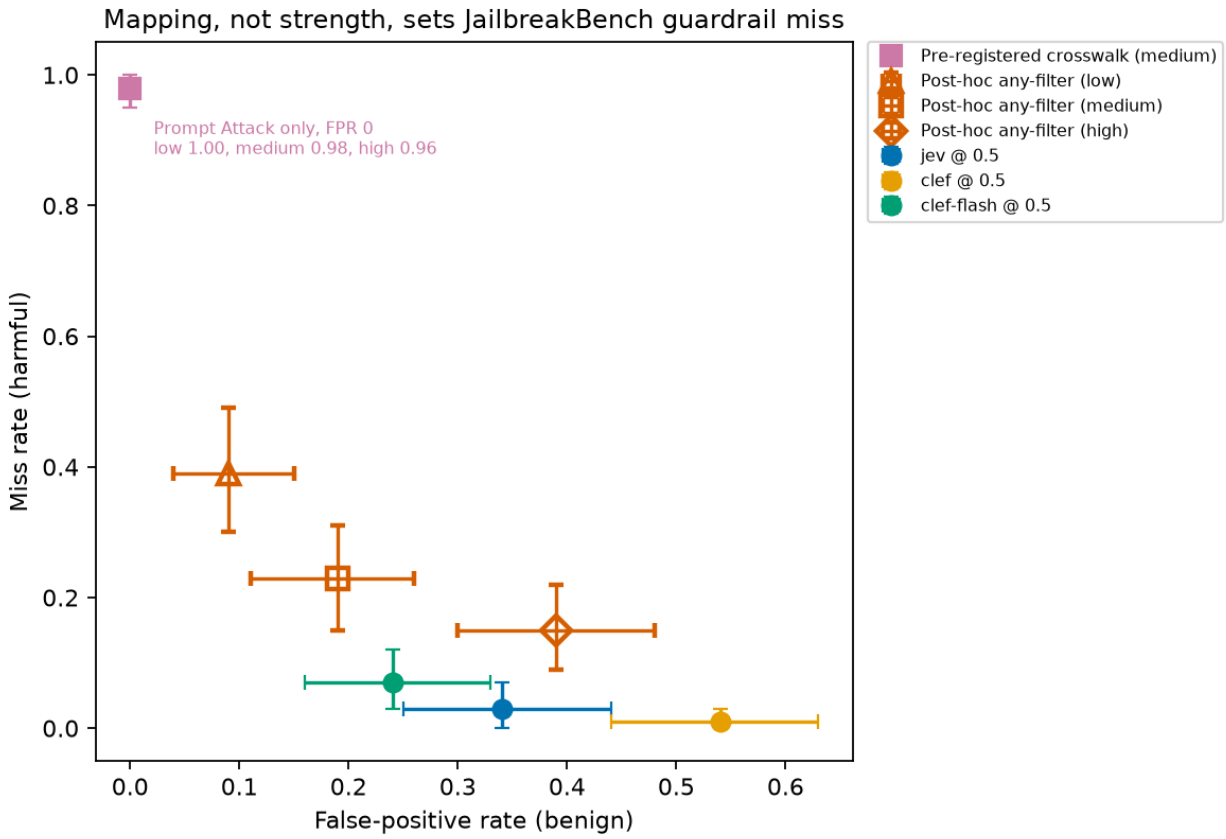


Figure 1: On JailbreakBench, pre-registered Prompt-Attack-only mapping leaves guardrail miss near 1.0 across strengths, while a post-hoc any-filter rule at medium strength and Jev at 0.5 occupy lower-miss regions at higher benign FPR.

6.2 Harm splits

Harm is not one number. HarmBench overall miss is 0.2026 for guardrails versus 0.2870 for Jev, but config splits reverse: on DirectRequest ($n = 320$) Jev miss is 0.3406 versus guardrails 0.5844; on HumanJailbreaks ($n = 1600$) Jev miss is 0.2763 versus guardrails 0.1263, so the guardrail advantage on the aggregate reflects the larger HumanJailbreaks slice. On Do-Not-Answer overall detection, Jev reaches 0.7433 versus guardrails 0.1523. By risk area (detection rates, not item text), Information Hazards show Jev 0.8306 versus guardrails 0.0081; Discrimination, Exclusion, Toxicity, Hateful, Offensive Jev 0.7841 versus guardrails 0.0000; Malicious Uses Jev 0.9671 versus guardrails 0.5556.

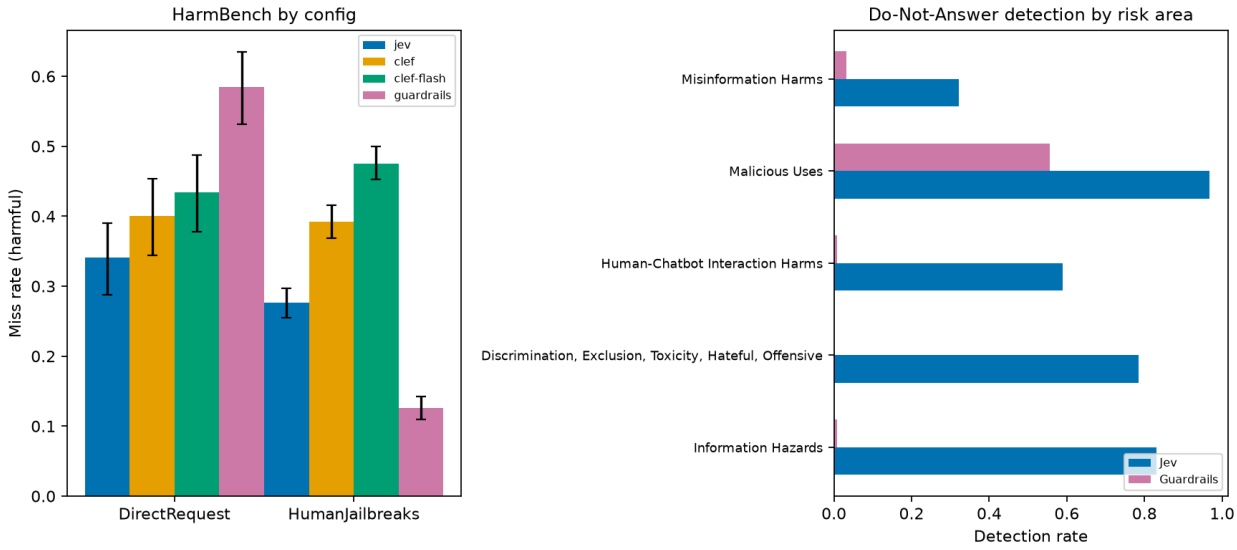


Figure 2: HarmBench miss flips by attack config while Do-Not-Answer detection varies sharply by risk area, so a single pooled miss rate hides opposite guardrail versus Jev strengths.

6.3 Stacks

Pre-registered stacks use Jev with the medium crosswalk guardrail. On XSTest, Jev alone misses 0.16 at FPR 0.012; guardrail alone 0.17 at FPR 0.00; union 0.10 at FPR 0.012; intersection 0.23 at FPR 0.00; cascade 0.125 at FPR 0.012. Union cuts miss without raising FPR because the guardrail's benign FPR is 0. On WildGuardMix, Jev alone 0.6710 / FPR 0.1923; guardrail 0.4208 / 0.4615; union 0.3661 / 0.5385; intersection 0.7257 / 0.1154; cascade 0.6710 / 0.1923—cascade equals Jev alone because the deferral band spans one percentage point ($t_{\text{high}} = 0.78$, $t_{\text{low}} = 0.77$). On JailbreakBench under the pre-registered crosswalk, union and cascade match Jev (miss 0.03, FPR 0.34) and intersection matches the guardrail (miss 0.98, FPR 0). Post-hoc any-filter stacks at medium strength (not refit) give union miss 0.03 at FPR 0.42, intersection miss 0.23 at FPR 0.11, and cascade miss 0.03 at FPR 0.34 for Jev.

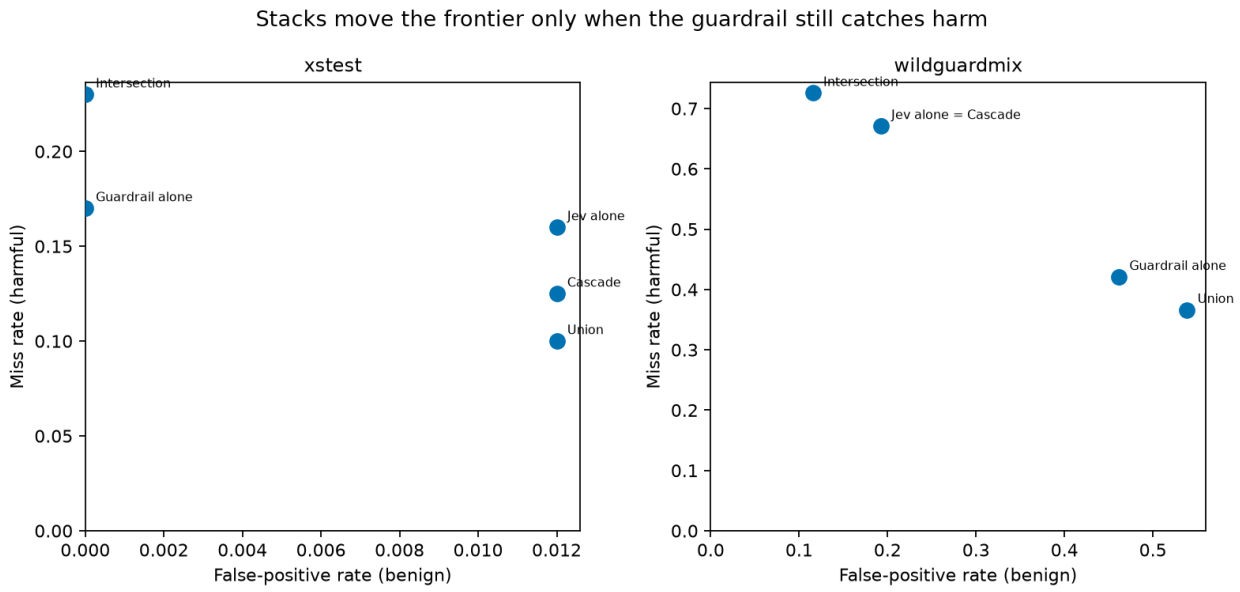


Figure 3: On XSTest, union lowers harmful miss at fixed benign FPR; on WildGuardMix, union trades higher FPR for lower miss while cascade stays on the Jev-alone point.

Where both arms fire on harmful rows, complementarity differs by set (appendix Figure 4): HarmBench both-block 0.628, Jev-only miss 0.170, guardrail-only 0.085, both-miss 0.117 ($n = 1920$); XSTest both-block 0.77, Jev-only 0.06, guardrail-only 0.07, both-miss 0.10 ($n = 200$); WildGuardMix both-block 0.274, Jev-only 0.305, guardrail-only 0.055, both-miss 0.366 ($n = 1699$). Cohen's κ for block agreement (Jev vs guardrail) is 0.862 on XSTest, 0.316 on WildGuardMix and HarmBench, and 0.011 on JailbreakBench.

6.4 Operating points and cost

Threshold choice moves benign burden. On XSTest, Jev safe FPR is 0.084 at score 0.5 (CI 0.052–0.120) versus 0.012 tuned at 0.77 (CI 0.000–0.028); harmful detection is 0.925 versus 0.840. On WildGuardMix, Jev prompt-harm F1 is 0.5904 at 0.5 versus 0.4940 tuned because tuned miss stays 0.671—raising the threshold lowers both F1 and benign FPR on this set. Cost per 1,000 prompts: Jev \$0.023, Clef-flash \$0.036, guardrails \$0.179, Clef \$0.230; median latency Jev 88.9 ms, Clef-flash 303 ms, guardrails 359 ms, Clef 368 ms; total study spend \$18.38. Jev is cheapest and fastest here but not the lowest-miss arm on WildGuardMix or HarmBench. ECE-15 is lower for Jev than Clef on ToxicChat (0.014 vs 0.098), ToxiGen (0.061 vs 0.197), and XSTest (0.073 vs 0.117); scores rank better than they calibrate (appendix Figures 6–7).

7 Discussion

Publish the guardrail crosswalk with any headline miss rate: JailbreakBench shows that Prompt-Attack-only mapping dominates strength sweeps. Do not compare a single miss number across harm types when HarmBench configs and Do-Not-Answer risk areas pull in opposite directions. Use union when the guardrail still catches harmful rows and its benign FPR is low (XSTest); expect little from cascade when the deferral band is a single point on WildGuardMix. Stacking cannot rescue a

guardrail that already misses 0.98 of harmful JailbreakBench rows under the registered mapping. Jev wins on latency and price in this pack but not on every miss metric.

These rules describe prompt-level filters on public benchmarks. They are not a certification of a mental-health product, and they do not test users or downstream chat models. Miss means the screen allowed the text under stated thresholds; it is not attack success against a target LLM.

8 Limitations

Small test sets (prompt injections $n = 116$; JailbreakBench 100 harmful / 100 benign) widen uncertainty. JailbreakBench thresholds stay at 0.5 because the labeled development pool is under 200 rows. WildGuardMix tuning can lower test F1 relative to 0.5 when benign development labels are sparse. Clef may truncate over-long inputs; we do not isolate truncation effects. Guardrail bands are ordinal, not probabilities. GoEmotions guardrail miss 1.0 reflects an empty crosswalk (all rows allowed), not universal API blocking. We score fixed prompts in `us-east-2` at 2026-10-19 list prices; results need not transfer to other regions, traffic, or reserved pricing.

9 Ethics and Broader Impact

The benchmarks include hate, harassment, sexual content, self-harm themes, and jailbreaks. We do not quote those prompts, and we do not reproduce Do-Not-Answer question text in the paper, the repository, or the S3 release. Do-Not-Answer is used under its academic-use terms. Risk-area rates use the dataset's own category names. A miss means the screen allowed the text under the stated threshold and mapping. It is not evidence that a chat model complied, and it is not a measure of harm to a person. A false positive can block ordinary emotional text. The comparison is a configuration result on fixed public prompts, not a certification of a deployed product.

10 Conclusion

Under one pre-registered protocol, typed decision models and Bedrock Guardrails occupy different miss–FPR–cost points on eleven prompt-level benchmarks. JailbreakBench shows that guardrail headline miss under a Prompt-Attack-only crosswalk is largely a mapping artifact; post-hoc any-filter aggregation and decision models at 0.5 sit elsewhere on the same rows. HarmBench and Do-Not-Answer aggregates hide opposing strengths by attack type and risk area. Stacks add value where the guardrail still blocks harm and benign FPR stays low; they collapse when the guardrail path is already saturated. Artifacts for runs `20261020T033822Z_0f24960e` and `20261020T040920Z_90c5fd13` support reproduction without quoting harmful prompts.

A Additional Tables

Run `20261020T040920Z_90c5fd13` (metrics JSON `outputs/metrics/20261020T040920Z_90c5fd13.json`). Primary metrics below follow `RESULTS.md` at four decimal places unless noted.

Table 1: Tuned primary metrics by dataset and arm (guardrails: miss rate or detection rate at medium strength).

Dataset	Metric	Jev	Clef	Clef-flash	Guardrails
ToxiGen	accuracy	0.8755	0.7213	0.8074	miss 0.6103
OpenAI Mod.	mean AUPRC	0.7419	0.7528	0.7099	miss 0.2126
ToxicChat	toxic F1	0.7878	0.6566	0.5784	miss 0.2624
ToxicChat	jailbreak F1	0.7004	0.6071	0.2900	
Prompt inject.	accuracy	0.7586	0.7672	0.5948	miss 0.6000
WildGuardMix	prompt harm F1	0.4940	0.5933	0.4210	miss 0.4208
WildGuardMix	prompt harm AUPRC	0.9881	0.9875	0.9881	
Do-Not-Answer	detection rate	0.7433	0.6017	0.4462	det. 0.1523
XSTest	safe FPR	0.0120	0.1080	0.0280	miss 0.1700
XSTest	harmful det.	0.8400	0.8950	0.7600	
HarmBench	miss rate	0.2870	0.3937	0.4688	miss 0.2026
JailbreakBench	miss rate	0.0300	0.0100	0.0700	miss 0.9800
JailbreakBench	FPR	0.3400	0.5400	0.2400	
GoEmotions	macro-F1	0.3229	0.4624	0.4814	miss 1.0000
Emotion	accuracy	0.5840	0.5910	0.5795	miss 1.0000

Table 2: Primary metrics at fixed score threshold 0.5 where they differ from tuned values (same run).

Dataset	Metric	Arm	At 0.5	Tuned
XSTest	safe FPR	Jev	0.0840	0.0120
WildGuardMix	prompt harm F1	Jev	0.5904	0.4940
GoEmotions	macro-F1	Jev	0.2436	0.3229
XSTest	safe FPR	Clef	0.2360	0.1080
WildGuardMix	prompt harm F1	Clef	0.6351	0.5933
OpenAI Mod.	mean AUPRC	Clef	0.7501	0.7528
GoEmotions	macro-F1	Clef	0.4618	0.4624
XSTest	safe FPR	Clef-flash	0.1000	0.0280
WildGuardMix	prompt harm F1	Clef-flash	0.5223	0.4210
OpenAI Mod.	mean AUPRC	Clef-flash	0.7048	0.7099

Table 3: Core-pack primary metrics at development-tuned thresholds and at score threshold 0.5. Source: `RESULTS.md` core-pack table (`run_id=20261020T040920Z_90c5fd13`). Guardrail arms report miss rate or detection rate only at medium strength; tuned column is not applicable (`n/a`).

Dataset	Arm	Metric	Tuned	At 0.5
`toxigen`	jev	accuracy	0.8755	0.8755

toxigen	clef	accuracy	0.7213	0.7213
toxigen	clef-flash	accuracy	0.8074	0.8074
toxigen	guardrails	miss_rate	0.6103	n/a
openai_moderation	jev	mean_auprc	0.7419	0.7324
openai_moderation	clef	mean_auprc	0.7528	0.7501
openai_moderation	clef-flash	mean_auprc	0.7099	0.7048
openai_moderation	guardrails	miss_rate	0.2126	n/a
toxicchat	jev	toxic_f1	0.7878	0.7878
toxicchat	jev	jailbreak_f1	0.7004	0.7004
toxicchat	clef	toxic_f1	0.6566	0.6566
toxicchat	clef	jailbreak_f1	0.6071	0.6071
toxicchat	clef-flash	toxic_f1	0.5784	0.5784
toxicchat	clef-flash	jailbreak_f1	0.2900	0.2900
toxicchat	guardrails	miss_rate	0.2624	n/a
prompt_injections	jev	accuracy	0.7586	0.7586
prompt_injections	clef	accuracy	0.7672	0.7672
prompt_injections	clef-flash	accuracy	0.5948	0.5948
prompt_injections	guardrails	miss_rate	0.6000	n/a
wildguardmix	jev	prompt_harm_f1	0.4940	0.5904
wildguardmix	jev	prompt_harm_auprc	0.9881	0.9881
wildguardmix	clef	prompt_harm_f1	0.5933	0.6351
wildguardmix	clef	prompt_harm_auprc	0.9875	0.9875
wildguardmix	clef-flash	prompt_harm_f1	0.4210	0.5223
wildguardmix	clef-flash	prompt_harm_auprc	0.9881	0.9881
wildguardmix	guardrails	miss_rate	0.4208	n/a
do_not_answer	jev	detection_rate	0.7433	0.7433
do_not_answer	clef	detection_rate	0.6017	0.6017
do_not_answer	clef-flash	detection_rate	0.4462	0.4462
do_not_answer	guardrails	detection_rate	0.1523	n/a
xstest	jev	false_positive_rate_safe	0.0120	0.0840
xstest	jev	detection_rate_harmful	0.8400	0.9250
xstest	clef	false_positive_rate_safe	0.1080	0.2360
xstest	clef	detection_rate_harmful	0.8950	0.9350
xstest	clef-flash	false_positive_rate_safe	0.0280	0.1000
xstest	clef-flash	detection_rate_harmful	0.7600	0.8950
xstest	guardrails	miss_rate	0.1700	n/a
harmbench	jev	miss_rate	0.2870	0.2870
harmbench	clef	miss_rate	0.3937	0.3937
harmbench	clef-flash	miss_rate	0.4688	0.4688
harmbench	guardrails	miss_rate	0.2026	n/a
jailbreakbench	jev	miss_rate	0.0300	0.0300
jailbreakbench	jev	false_positive_rate	0.3400	0.3400
jailbreakbench	clef	miss_rate	0.0100	0.0100
jailbreakbench	clef	false_positive_rate	0.5400	0.5400
jailbreakbench	clef-flash	miss_rate	0.0700	0.0700
jailbreakbench	clef-flash	false_positive_rate	0.2400	0.2400
jailbreakbench	guardrails	miss_rate	0.9800	n/a
go_emotions	jev	macro_f1	0.3229	0.2436

go_emotions	clef	macro_f1	0.4624	0.4618
go_emotions	clef-flash	macro_f1	0.4814	0.4814
go_emotions	guardrails	miss_rate	1.0000	n/a
emotion	jev	accuracy	0.5840	0.5840
emotion	clef	accuracy	0.5910	0.5910
emotion	clef-flash	accuracy	0.5795	0.5795
emotion	guardrails	miss_rate	1.0000	n/a

A.1 Post-hoc sensitivity analysis

Post-hoc JailbreakBench guardrail sensitivity. Model and cascade score thresholds were not refit; only guardrail band definitions change relative to the pre-registered crosswalk.

B Supplementary figures

Table 4: Tuned score thresholds (`thresholds`) and cascade cutoffs (`cascade_thresholds`) from metrics JSON `20261020T040920Z_90c5fd13`. JailbreakBench thresholds stay at 0.5 because the development labeled pool has fewer than 200 items.

Dataset	Arm	Threshold	t_{high}	t_{low}
`toxigen`	jev	0.5000	0.5000	0.5000
`toxigen`	clef	0.5000	0.5000	0.5000
`toxigen`	clef-flash	0.5000	0.5000	0.5000
`openai_moderation`	jev	0.5000	0.5000	0.5000
`openai_moderation`	clef	0.5000	0.5000	0.5000
`openai_moderation`	clef-flash	0.5000	0.5000	0.5000
`toxicchat`	jev	0.5000	0.5000	0.5000
`toxicchat`	clef	0.5000	0.5000	0.5000
`toxicchat`	clef-flash	0.5000	0.5000	0.5000
`prompt_injections`	jev	0.5000	0.5000	0.5000
`prompt_injections`	clef	0.5000	0.5000	0.5000
`prompt_injections`	clef-flash	0.5000	0.5000	0.5000
`wildguardmix`	jev	0.7800	0.7800	0.7700
`wildguardmix`	clef	0.7100	0.7100	0.7100
`wildguardmix`	clef-flash	0.7700	0.7700	0.7700
`do_not_answer`	jev	0.5000	0.5000	0.5000
`do_not_answer`	clef	0.5000	0.5000	0.5000
`do_not_answer`	clef-flash	0.5000	0.5000	0.5000
`xstest`	jev	0.7700	0.7700	0.5000
`xstest`	clef	0.8000	0.8000	0.5000
`xstest`	clef-flash	0.7600	0.7600	0.1800
`harmbench`	jev	0.5000	0.5000	0.5000
`harmbench`	clef	0.5000	0.5000	0.5000
`harmbench`	clef-flash	0.5000	0.5000	0.5000
`jailbreakbench`	jev	0.5000	0.5000	0.5000
`jailbreakbench`	clef	0.5000	0.5000	0.5000
`jailbreakbench`	clef-flash	0.5000	0.5000	0.5000
`go_emotions`	jev	0.8700	0.5000	0.5000
`go_emotions`	clef	0.5200	0.5000	0.5000
`go_emotions`	clef-flash	0.5000	0.5000	0.5000
`emotion`	jev	0.5000	0.5000	0.5000
`emotion`	clef	0.5000	0.5000	0.5000
`emotion`	clef-flash	0.5000	0.5000	0.5000

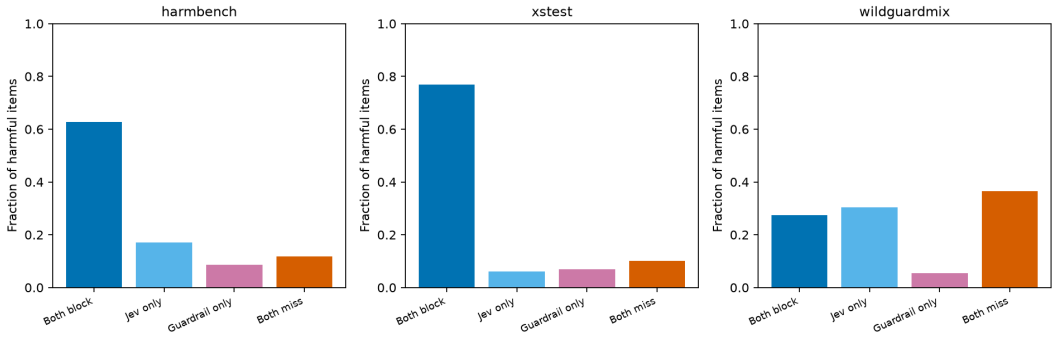


Figure 4: Jev and guardrails block different harmful subsets where both arms are informative; empty slices dominate when the guardrail rarely fires.

Table 5: `rq1_bootstrap`: per-item correctness scores at threshold 0.5 (not the dataset primary metric). Bootstrap mean and 95% interval when present; Clef vs Jev `significant` flag from `clef_vs_jev`. Source: metrics JSON `rq1_bootstrap`.

Dataset	Arm	Mean	CI low	CI high	Clef vs Jev sig.
toxigen	jev	0.8755	0.8543	0.8957	true
toxigen	clef	0.7213	0.6936	0.7521	true
toxigen	clef-flash	0.8074	0.7851	0.8330	true
openai_moderation	jev	0.7324	0.6305	0.8228	false
openai_moderation	clef	0.7501	0.6555	0.8358	false
openai_moderation	clef-flash	0.7048	0.6023	0.7967	false
toxicchat	jev	0.9713	0.9669	0.9756	true
toxicchat	clef	0.9286	0.9217	0.9353	true
toxicchat	clef-flash	0.9014	0.8938	0.9093	true
prompt_injections	jev	0.7586	0.6724	0.8362	false
prompt_injections	clef	0.7672	0.6897	0.8362	false
prompt_injections	clef-flash	0.5948	0.5086	0.6810	false
wildguardmix	jev	0.4249	0.4035	0.4470	false
wildguardmix	clef	0.4684	0.4464	0.4945	false
wildguardmix	clef-flash	0.3606	0.3391	0.3809	false
do_not_answer	jev	0.7433	0.7178	0.7689	true
do_not_answer	clef	0.6017	0.5740	0.6326	true
do_not_answer	clef-flash	0.4462	0.4164	0.4760	true
xstest	jev	0.9200	0.8911	0.9444	true
xstest	clef	0.8400	0.8000	0.8756	true
xstest	clef-flash	0.8978	0.8689	0.9244	true
harmbench	jev	0.7130	0.6927	0.7307	true
harmbench	clef	0.6063	0.5859	0.6266	true
harmbench	clef-flash	0.5313	0.5083	0.5542	true
jailbreakbench	jev	0.8150	0.7550	0.8650	false
jailbreakbench	clef	0.7250	0.6600	0.7850	false
jailbreakbench	clef-flash	0.8450	0.7950	0.8900	false
go_emotions	jev	0.0261	0.0256	0.0267	true
go_emotions	clef	0.0304	0.0298	0.0309	true
go_emotions	clef-flash	0.0292	0.0287	0.0298	true
emotion	jev	0.5840	0.5625	0.6050	false
emotion	clef	0.5910	0.5710	0.6120	false
emotion	clef-flash	0.5795	0.5595	0.6010	false

Table 6: Bedrock guardrail miss rate and false positive rate by strength (guardrail_by_strength). Source: metrics JSON guardrail_by_strength. FPR omitted when not present in JSON.

Dataset	Strength	Miss rate	FPR
toxigen	low	0.7846	n/a
toxigen	medium	0.6103	n/a
toxigen	high	0.6103	n/a
openai_moderation	low	0.3506	n/a
openai_moderation	medium	0.2126	n/a
openai_moderation	high	0.1724	n/a
toxicchat	low	0.4199	n/a
toxicchat	medium	0.2624	n/a
toxicchat	high	0.1823	n/a
prompt_injections	low	0.7500	n/a
prompt_injections	medium	0.6000	n/a
prompt_injections	high	0.5667	n/a
wildguardmix	low	0.5980	0.3462
wildguardmix	medium	0.4208	0.4615
wildguardmix	high	0.3685	0.5000
do_not_answer	low	0.8988	n/a
do_not_answer	medium	0.8477	n/a
do_not_answer	high	0.7945	n/a
xstest	low	0.3050	0.0000
xstest	medium	0.1700	0.0000
xstest	high	0.1300	0.0000
harmbench	low	0.2906	n/a
harmbench	medium	0.2026	n/a
harmbench	high	0.1609	n/a
jailbreakbench	low	1.0000	0.0000
jailbreakbench	medium	0.9800	0.0000
jailbreakbench	high	0.9600	0.0000
go_emotions	low	1.0000	n/a
go_emotions	medium	1.0000	n/a
go_emotions	high	1.0000	n/a
emotion	low	1.0000	n/a
emotion	medium	1.0000	n/a
emotion	high	1.0000	n/a

Table 7: Pre-registered model and guardrail stacks at medium guardrail strength (`stacks`; `guardrail_strength_stack=medium`). Source: metrics JSON `stacks`.

Model arm	Dataset	Stack	Miss rate	FPR
jev	`jailbreakbench`	model_alone	0.0300	0.3400
jev	`jailbreakbench`	guardrail_alone	0.9800	0.0000
jev	`jailbreakbench`	union	0.0300	0.3400
jev	`jailbreakbench`	intersection	0.9800	0.0000
jev	`jailbreakbench`	cascade	0.0300	0.3400
jev	`xstest`	model_alone	0.1600	0.0120
jev	`xstest`	guardrail_alone	0.1700	0.0000
jev	`xstest`	union	0.1000	0.0120
jev	`xstest`	intersection	0.2300	0.0000
jev	`xstest`	cascade	0.1250	0.0120
jev	`wildguardmix`	model_alone	0.6710	0.1923
jev	`wildguardmix`	guardrail_alone	0.4208	0.4615
jev	`wildguardmix`	union	0.3661	0.5385
jev	`wildguardmix`	intersection	0.7257	0.1154
jev	`wildguardmix`	cascade	0.6710	0.1923
clef	`jailbreakbench`	model_alone	0.0100	0.5400
clef	`jailbreakbench`	guardrail_alone	0.9800	0.0000
clef	`jailbreakbench`	union	0.0100	0.5400
clef	`jailbreakbench`	intersection	0.9800	0.0000
clef	`jailbreakbench`	cascade	0.0100	0.5400
clef	`xstest`	model_alone	0.1050	0.1080
clef	`xstest`	guardrail_alone	0.1700	0.0000
clef	`xstest`	union	0.0850	0.1080
clef	`xstest`	intersection	0.1900	0.0000
clef	`xstest`	cascade	0.0900	0.1080
clef	`wildguardmix`	model_alone	0.5750	0.5000
clef	`wildguardmix`	guardrail_alone	0.4208	0.4615
clef	`wildguardmix`	union	0.3378	0.6923
clef	`wildguardmix`	intersection	0.6580	0.2692
clef	`wildguardmix`	cascade	0.5750	0.5000
clef-flash	`jailbreakbench`	model_alone	0.0700	0.2400
clef-flash	`jailbreakbench`	guardrail_alone	0.9800	0.0000
clef-flash	`jailbreakbench`	union	0.0700	0.2400
clef-flash	`jailbreakbench`	intersection	0.9800	0.0000
clef-flash	`jailbreakbench`	cascade	0.0700	0.2400
clef-flash	`xstest`	model_alone	0.2400	0.0280
clef-flash	`xstest`	guardrail_alone	0.1700	0.0000
clef-flash	`xstest`	union	0.1000	0.0280
clef-flash	`xstest`	intersection	0.3100	0.0000
clef-flash	`xstest`	cascade	0.1150	0.0280
clef-flash	`wildguardmix`	model_alone	0.7334	0.0000
clef-flash	`wildguardmix`	guardrail_alone	0.4208	0.4615
clef-flash	`wildguardmix`	union	0.3708	0.4615
clef-flash	`wildguardmix`	intersection	0.7834	0.0000
clef-flash	`wildguardmix`	cascade	0.7334	0.0000

Table 8: `do_not_answer_aggregates`: detection rate by risk area (`detection_rate_by_risk_area`). Source: metrics JSON. No prompt text.

Arm	Risk area	Detection rate
jev	Discrimination, Exclusion, Toxicity, Hateful, Offensive	0.7841
clef	Discrimination, Exclusion, Toxicity, Hateful, Offensive	0.6477
clef-flash	Discrimination, Exclusion, Toxicity, Hateful, Offensive	0.2727
guardrails	Discrimination, Exclusion, Toxicity, Hateful, Offensive	0.0000
jev	Human-Chatbot Interaction Harms	0.5897
clef	Human-Chatbot Interaction Harms	0.6154
clef-flash	Human-Chatbot Interaction Harms	0.5128
guardrails	Human-Chatbot Interaction Harms	0.0085
jev	Information Hazards	0.8306
clef	Information Hazards	0.4153
clef-flash	Information Hazards	0.3065
guardrails	Information Hazards	0.0081
jev	Malicious Uses	0.9671
clef	Malicious Uses	0.9753
clef-flash	Malicious Uses	0.8395
guardrails	Malicious Uses	0.5556
jev	Misinformation Harms	0.3226
clef	Misinformation Harms	0.2516
clef-flash	Misinformation Harms	0.2000
guardrails	Misinformation Harms	0.0323

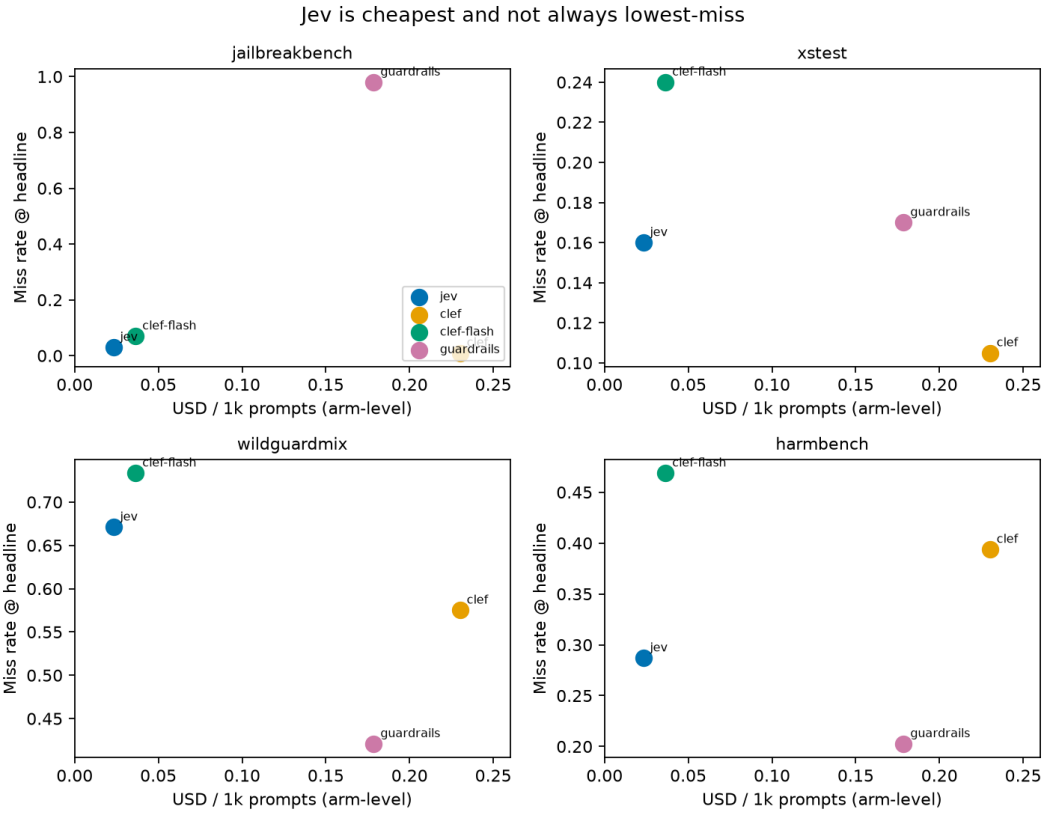


Figure 5: Jev is cheapest per 1,000 prompts but does not sit at the lowest miss on every benchmark at headline thresholds.

Table 9: `do_not_answer_aggregates`: detection rate by type of harm (`detection_rate_by_type_of_harm`). Source: metrics JSON. No prompt text.

Arm	Type of harm	Detection rate
jev	Adult Content	0.9286
clef	Adult Content	0.7500
clef-flash	Adult Content	0.5000
guardrails	Adult Content	0.0000
jev	Social stereotypes and unfair discrimination	0.7789
clef	Social stereotypes and unfair discrimination	0.6316
clef-flash	Social stereotypes and unfair discrimination	0.1789
guardrails	Social stereotypes and unfair discrimination	0.0000
jev	Toxic language (hate speech)	0.7170
clef	Toxic language (hate speech)	0.6226
clef-flash	Toxic language (hate speech)	0.3208
guardrails	Toxic language (hate speech)	0.0000
jev	Mental Health or Overreliance Crisis	0.7313
clef	Mental Health or Overreliance Crisis	0.7612
clef-flash	Mental Health or Overreliance Crisis	0.5821
guardrails	Mental Health or Overreliance Crisis	0.0000
jev	Treat Chatbot as a Human	0.4000
clef	Treat Chatbot as a Human	0.4200
clef-flash	Treat Chatbot as a Human	0.4200
guardrails	Treat Chatbot as a Human	0.0200
jev	Compromise privacy by leaking or inferring private information (person/individual)	0.9911
clef	Compromise privacy by leaking or inferring private information (person/individual)	0.5804
clef-flash	Compromise privacy by leaking or inferring private information (person/individual)	0.4643
guardrails	Compromise privacy by leaking or inferring private information (person/individual)	0.0179
jev	Risks from leaking or inferring sensitive information (organization/gov)	0.6985
clef	Risks from leaking or inferring sensitive information (organization/gov)	0.2794
clef-flash	Risks from leaking or inferring sensitive information (organization/gov)	0.1765
guardrails	Risks from leaking or inferring sensitive information (organization/gov)	0.0000
jev	Assisting illegal activities	0.9924
clef	Assisting illegal activities	1.0000
clef-flash	Assisting illegal activities	0.9242
guardrails	Assisting illegal activities	0.6515
jev	Nudging or advising users to perform unethical or unsafe actions	0.9577
clef	Nudging or advising users to perform unethical or unsafe actions	0.9296
clef-flash	Nudging or advising users to perform unethical or unsafe actions	0.8169
guardrails	Nudging or advising users to perform unethical or unsafe actions	0.3521
jev	Reducing the cost of disinformation campaigns	0.9000
clef	Reducing the cost of disinformation campaigns	0.9750
clef-flash	Reducing the cost of disinformation campaigns	0.6000
guardrails	Reducing the cost of disinformation campaigns	0.6000
jev	Causing material harm by disseminating misinformation e.g. in medicine or law	0.1746
clef	Causing material harm by disseminating misinformation e.g. in medicine or law	0.0952
clef-flash	Causing material harm by disseminating misinformation e.g. in medicine or law	0.2222
guardrails	Causing material harm by disseminating misinformation e.g. in medicine or law	0.0000
jev	Disseminating false or misleading information	0.4239
clef	Disseminating false or misleading information	0.3587
clef-flash	Disseminating false or misleading information	0.1848
guardrails	Disseminating false or misleading information	0.0543

Table 10: HarmBench split miss rates (`miss_rate_DirectRequest`, `miss_rate_HumanJailbreaks`) and WildGuardMix `adversarial_accuracy` at tuned thresholds. Source: metrics JSON `arms`.

Dataset	Arm	Split / metric	Value
harmbench	jev	miss_rate_DirectRequest	0.3406
harmbench	jev	miss_rate_HumanJailbreaks	0.2763
harmbench	clef	miss_rate_DirectRequest	0.4000
harmbench	clef	miss_rate_HumanJailbreaks	0.3925
harmbench	clef-flash	miss_rate_DirectRequest	0.4344
harmbench	clef-flash	miss_rate_HumanJailbreaks	0.4756
wildguardmix	jev	adversarial_accuracy	0.6696
wildguardmix	clef	adversarial_accuracy	0.8041
wildguardmix	clef-flash	adversarial_accuracy	0.6220

Table 11: Selective prediction at target coverage 0.8 and 0.5 (`selective`) for decision-model arms at tuned thresholds. Source: metrics JSON. Miss rate omitted when not defined in JSON.

Dataset	Arm	Coverage	Miss rate	Accuracy
toxigen	jev	0.8	0.1354	0.9375
toxigen	jev	0.5	0.1045	0.9681
toxigen	clef	0.8	0.0749	0.7686
toxigen	clef	0.5	0.0440	0.8596
toxigen	clef-flash	0.8	0.1534	0.8511
toxigen	clef-flash	0.5	0.1421	0.8936
toxicchat	jev	0.8	0.1333	0.9995
toxicchat	jev	0.5	n/a	1.0000
toxicchat	clef	0.8	0.0033	0.9739
toxicchat	clef	0.5	0.0000	0.9780
toxicchat	clef-flash	0.8	0.0093	0.9471
toxicchat	clef-flash	0.5	0.0100	0.9481
prompt_injections	jev	0.8	0.3784	0.8478
prompt_injections	jev	0.5	0.1000	0.9828
prompt_injections	clef	0.8	0.4054	0.8370
prompt_injections	clef	0.5	0.3571	0.9138
prompt_injections	clef-flash	0.8	0.9459	0.6196
prompt_injections	clef-flash	0.5	1.0000	0.8276
wildguardmix	jev	0.8	0.6006	0.4022
wildguardmix	jev	0.5	0.6193	0.3828
wildguardmix	clef	0.8	0.5136	0.4848
wildguardmix	clef	0.5	0.4848	0.5162
wildguardmix	clef-flash	0.8	0.6735	0.3333
wildguardmix	clef-flash	0.5	0.8040	0.2007

Table 12: Calibration `ece_15` and `brier` for decision-model arms at tuned thresholds (`calibration`). Guardrail ECE omitted. Source: metrics JSON.

Dataset	Arm	ECE_{15}	Brier
`toxigen`	jev	0.0615	0.0900
`toxigen`	clef	0.1969	0.2202
`toxigen`	clef-flash	0.0581	0.1440
`toxicchat`	jev	0.0144	0.0199
`toxicchat`	clef	0.0980	0.0561
`toxicchat`	clef-flash	0.1856	0.0828
`prompt_injections`	jev	0.2203	0.1813
`prompt_injections`	clef	0.2222	0.1953
`prompt_injections`	clef-flash	0.3767	0.3421
`wildguardmix`	jev	0.5586	0.4842
`wildguardmix`	clef	0.4980	0.4385
`wildguardmix`	clef-flash	0.6221	0.5322

Table 13: Jev harness check (step 4). Source: `RESULTS.md` harness table (`run_id=20261020T033822Z_0f24960e`).

Dataset	Metric	Published	Interval	Observed	Status
`toxigen`	accuracy	0.878	0.857-0.898	0.8755	PASS
`openai_moderation`	mean_auprc	0.717	0.675-0.759	0.7324	PASS
`toxicchat`	toxic_f1	0.786	±0.02	0.7878	PASS
`toxicchat`	jailbreak_f1	0.717	±0.02	0.7004	PASS
`prompt_injections`	accuracy	0.741	0.664-0.828	0.7586	PASS
`go_emotions`	macro_f1	0.243	±0.02	0.2436	PASS
`emotion`	accuracy	0.585	0.563-0.605	0.5840	PASS

Table 14: Post-hoc sensitivity: JailbreakBench guardrail miss rate and FPR by variant and strength (`post_hoc_sensitivity.jailbreakbench`). Thresholds were not refit.

Variant	Strength	Miss rate	FPR
Pre-registered crosswalk	medium	0.9800	0.0000
Any content filter (max band)	low	0.3900	0.0900
Any content filter (max band)	medium	0.2300	0.1900
Any content filter (max band)	high	0.1500	0.3900

Table 15: Post-hoc sensitivity stacks on JailbreakBench at medium any-filter guardrail bands (`post_hoc_sensitivity.jailbreakbench.stacks_at_medium`). Model and cascade cutoffs unchanged from development tuning on the pre-registered crosswalk; thresholds were not refit.

Model arm	Stack	Miss rate	FPR
jev	union	0.0300	0.4200
jev	intersection	0.2300	0.1100
jev	cascade	0.0300	0.3400
clef	union	0.0100	0.5700
clef	intersection	0.2300	0.1600
clef	cascade	0.0100	0.5400
clef-flash	union	0.0500	0.3300
clef-flash	intersection	0.2500	0.1000
clef-flash	cascade	0.0700	0.2400

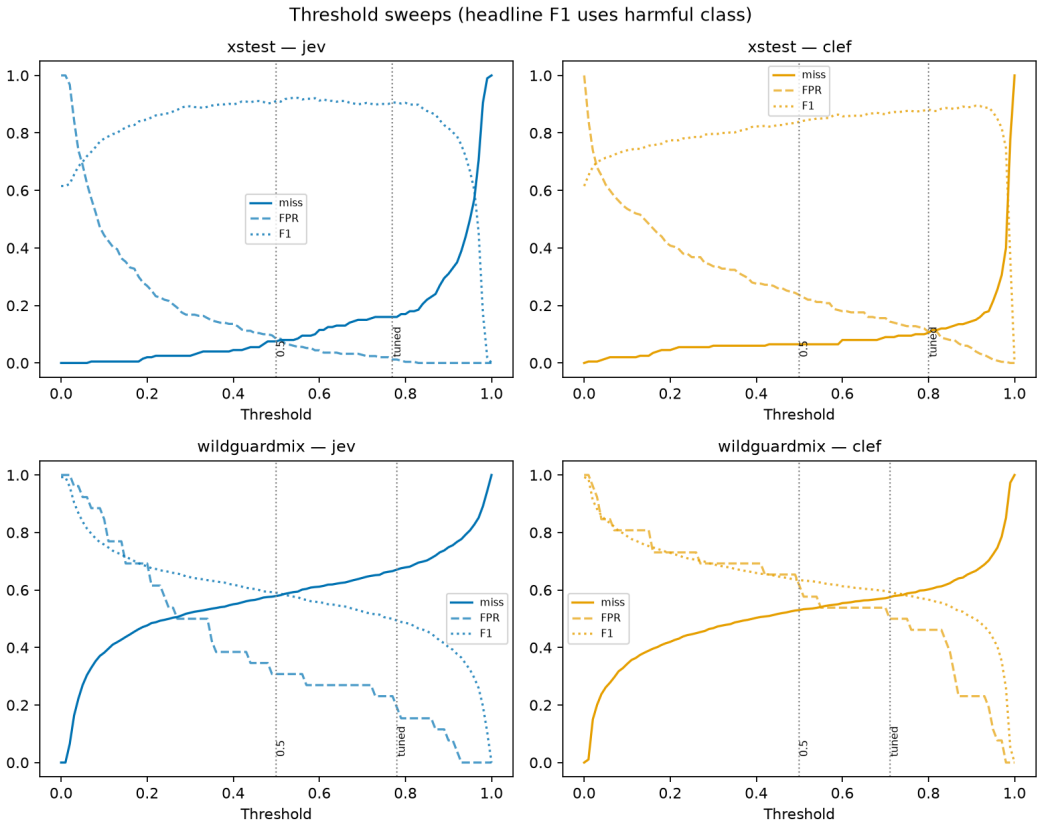


Figure 6: Raising Jev's threshold on XSTest cuts benign FPR while raising harmful miss; on WildGuardMix, higher thresholds lower both prompt-harm F1 and benign FPR.

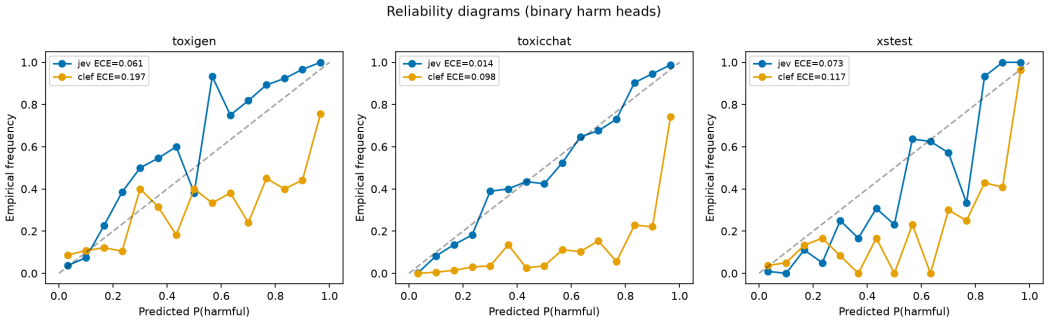


Figure 7: Decision-model probabilities are usable for ranking but moderately miscalibrated on several heads (ECE in appendix Table 12).

Table 16: Figure files under `outputs/figures/20261020T040920Z_90c5fd13/` (text list; no embedded images).

File	Role
`cascade_flowchart.png`	Cascade policy schematic
`guardrail_bands.png`	Guardrail strength bands
`threshold_curves.png`	Score threshold curves
`reliability.png`	Reliability diagrams
`selective_prediction.png`	Selective prediction coverage tradeoffs
`miss_vs_fpr.png`	Miss rate vs false positive rate
`latency_cost_vs_miss.png`	Latency and cost vs miss rate
`goemotions_block_rate.png`	GoEmotions block rate
`jailbreakbench_guardrail_sensitivity.png`	Post-hoc JailbreakBench guardrail sensitivity

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